

3. Equilibrium of a Particle

Tuesday, August 20, 2024 8:42 AM

* Equilibrium: @ rest or constant velocity
 $a=0$

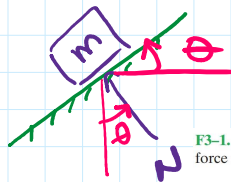
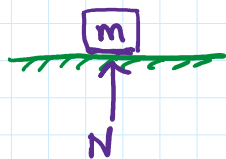
* $\sum \vec{F} = 0$

* Free-Body Diagram (FBD)
 of a point (particle)

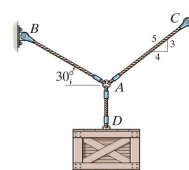
* Spring Force: $F = k \cdot \underline{s} = k(l - l_0)$
 $\downarrow$ stiffness constant $\left[\frac{N}{m}\right]$
 $\uparrow$ unstrained length

* cables in tension

* Normal Force



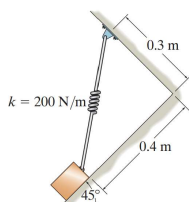
F3-1. The crate has a weight of 550 lb. Determine the force in each supporting cable.



Prob. F3-1

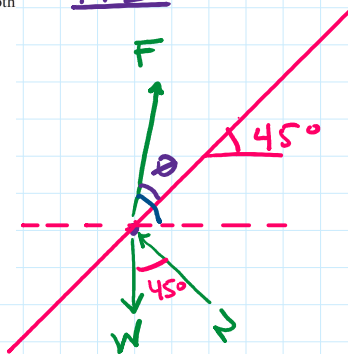
* 2D Space $\sum F_x = 0$, $\sum F_y = 0$
 $\sum F_t = 0$, $\sum F_n = 0$

F3-4. The block has a mass of 5 kg and rests on the smooth plane. Determine the unstretched length of the spring.



Prob. F3-4

FBD



$$\sum F_x = 0$$

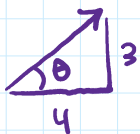
$$\sum F_y = 0$$

$$N_x = N \sin(45^\circ)$$

$$N_y = N \cos(45^\circ)$$

$$F_x = F \cos(\theta + 45^\circ)$$

$$F_y = F \sin(\theta + 45^\circ)$$



$$\tan(\theta) = \frac{3}{4}$$

$$\theta = \tan^{-1}\left(\frac{3}{4}\right) = 36.9^\circ \approx 37^\circ$$

$$\sum F_x = 0 \quad F_x - N_x = 0 \rightarrow F_x = N_x$$

$$\textcircled{1} \quad F \cos(82^\circ) = N \sin(45^\circ) \rightarrow \underline{1a} \quad N = \frac{F \cos(82^\circ)}{\sin(45^\circ)}$$

$$\sum F_y = 0 \quad F_y + N_y - W = 0$$

$$\sum F_x = 0 \quad F_y + N_y - W = 0$$

$$(2) \quad F \sin(82^\circ) + N \cos(45^\circ) = mg$$

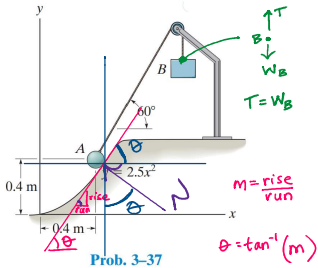
$$F \sin(82^\circ) + \frac{F \cos(82^\circ)}{\sin(45^\circ)} \cos(45^\circ) = mg$$

$$F (\sin(82^\circ) + \cos(82^\circ)) = mg$$

$$F = \frac{mg}{\sin(82^\circ) + \cos(82^\circ)} = \frac{(5 \text{ kg})(9.81 \text{ m/s}^2)}{\sin(82^\circ) + \cos(82^\circ)} = 43.41 \text{ N}$$

$$l_0 = l - \frac{F}{k} = 0.5 - \frac{43.41}{200} = 0.28 \text{ m}$$

3-37. A 4-kg sphere rests on the smooth parabolic surface. Determine the normal force it exerts on the surface and the mass m_B of block B needed to hold it in the equilibrium position shown.

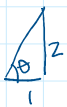


$$y = 2.5x^2$$

* to get slope of tangent line

$$\left. \frac{dy}{dx} \right|_{x=0.4} = 5x = 5(0.4) = 2$$

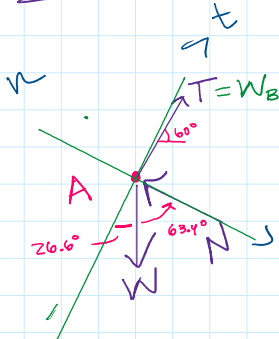
$$m = 2 = \frac{\text{rise}}{\text{run}}$$



$$\theta = \tan^{-1}\left(\frac{2}{1}\right)$$

$$\theta = 63.4^\circ$$

FBD



Equilibrium $\sum \vec{F} = 0$
 $\sum F_x = 0$, $\sum F_y = 0$

$$T_x = T \cos(60^\circ) = \frac{1}{2}T = 0.5T$$

$$T_y = T \sin(60^\circ) = \frac{\sqrt{3}}{2}T = 0.866T$$

$$N_x = -N \sin(63.4^\circ) = -0.89N$$

$$N_y = N \cos(63.4^\circ) = 0.44N$$

$$W_x = 0$$

$$W_y = W_A$$

$$\sum F_x = 0 \quad T_x - N_x = 0 \quad (1)$$

$$0.5T - 0.89N = 0$$

$$\sum F_y = 0 \quad T_y + N_y - W = 0 \quad (2)$$

$$0.866T + 0.44N = W_A \rightarrow m_A g$$

$$W_A = (4 \text{ kg})(9.81 \text{ m/s}^2) = 39.2 \text{ N}$$

$$(1a) \quad 0.5T - 0.89N = 0$$

$$0.5T = 0.89N$$

$$T = 1.78 \text{ N}$$

$$(2a)$$

$$0.866(1.78 \text{ N}) + 0.44N = 39.2$$

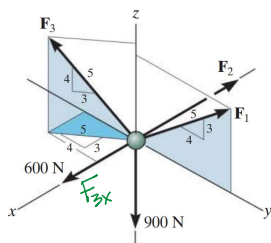
$$1.98 \text{ N} = 39.2$$

$$N = 19.78 \text{ N}$$

$$T = 1.78(19.78) = 35.21 \text{ N}$$

$$m_B = \frac{W_B}{g} = \frac{T}{g} = \frac{35.21 \text{ N}}{9.81 \text{ m/s}^2} = 3.58 \text{ kg}$$

F3-7. Determine the magnitude of forces F_1 , F_2 , F_3 , so that the particle is held in equilibrium.



Prob. F3-7

x, y, z components
of F_1, F_2, F_3

$$F_{1x} = 0 \quad F_{1y} = \frac{4}{5}F_1 \quad F_{1z} = \frac{3}{5}F_1$$

$$F_{2x} = -F_2 \quad F_{2y} = 0 \quad F_{2z} = 0$$

$$F_{3x} = \frac{9}{25}F_3 \quad F_{3y} = -\frac{12}{25}F_3 \quad F_{3z} = \frac{4}{5}F_3$$

$$\sum \vec{F} = 0 \quad \text{Equilibrium}$$

$$\sum F_x = 0 \quad \sum F_y = 0 \quad \sum F_z = 0$$

$$\sum F_x = 0$$

$$F_{1x} + F_{2x} + F_{3x} + 600 = 0$$

$$0 - F_2 + \frac{9}{25}F_3 = -600 \quad (1)$$

$$\sum F_y = 0$$

$$4/5 F_1 + 0 - 12/25 F_3 = 0 \quad (2)$$

$$\begin{bmatrix} 0 & -1 & 9/25 \\ 4/5 & 0 & -12/25 \end{bmatrix} \cdot \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix} = \begin{bmatrix} -600 \\ 0 \end{bmatrix}$$

$$\sum F_y = 0$$

$$4/5 F_1 + 0 - \frac{12}{25} F_3 = 0 \quad (2)$$

$$\sum F_z = 0$$

$$3/5 F_1 + 0 + 4/5 F_3 = 900 \quad (3)$$

$$F_{1z} + F_{2z} + F_{3z} - 900 = 0$$

$$\begin{bmatrix} 0 & 1 & 12/25 \\ 4/5 & 0 & -12/25 \\ 3/5 & 0 & 4/5 \end{bmatrix} \cdot \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix} = \begin{bmatrix} -600 \\ 0 \\ 900 \end{bmatrix}$$

$$x = A^{-1} b$$

$$(2a) \quad 4/5 F_1 - \frac{12}{25} F_3 = 0$$

$$F_1 = \left(\frac{3}{4}\right) \left(\frac{12}{25} F_3\right)$$

$$F_1 = \frac{3}{5} F_3 \rightarrow \text{substitute into (3)}$$

Roy

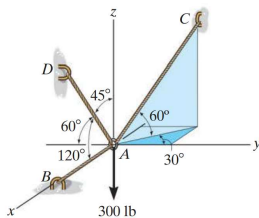
Oscar

Gabriel

Hannah

Ivan

F3-10. Determine the tension developed in cables AB, AC, and AD.



Prob. F3-10

$$AD_x = AD \cos(120^\circ) = -0.5 AD$$

$$AD_y = -AD \cos(60^\circ) = -0.5 AD$$

$$AD_z = AD \cos(45^\circ) = 0.707 AD$$

$$AC_x = AC \cos(60^\circ) \sin(30^\circ) = -0.25 AC$$

$$AC_y = AC \cos(60^\circ) \cos(30^\circ) = 0.433 AC$$

$$AC_z = AC \sin(60^\circ) = 0.866 AC$$

$$AB_x = AB$$

$$AB_y = 0$$

$$AB_z = 0$$

$$\sum F_x = 0$$

$$(1) \quad -0.5 AD - 0.25 AC + AB = 0$$

$$\sum F_y = 0$$

$$(2) \quad -0.5 AD + 0.433 AC + 0 = 0 \rightarrow$$

$$AD = 0.866 AC$$

Substitute

$$\sum F_z = 0$$

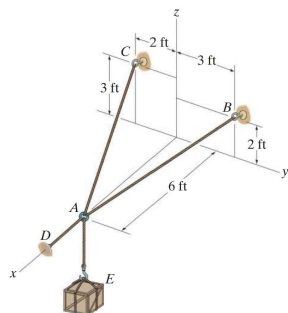
$$(3) \quad 0.707 AD + 0.866 AC + 0 = 300$$

$$AB = 139 \text{ lb}$$

$$AC = 203 \text{ lb}$$

$$AD = 176 \text{ lb}$$

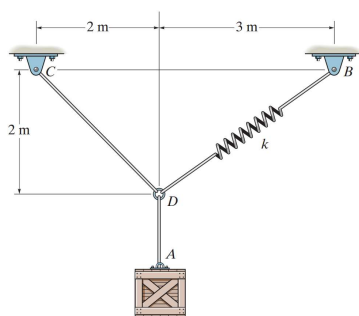
F3-11. The 150-lb crate is supported by cables AB , AC , and AD . Determine the tension in these wires.



Prob. F3-11

3-18. If the spring DB has an unstretched length of 2 m, determine the stiffness of the spring to hold the 40-kg crate in the position shown.

3-19. Determine the unstretched length of DB to hold the 40-kg crate in the position shown. Take $k = 180 \text{ N/m}$.



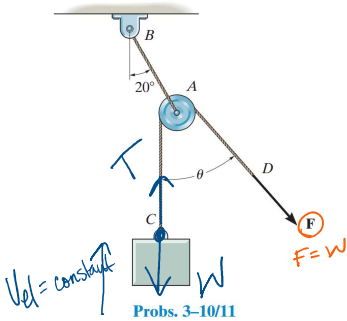
Probs. 3-18/19

3-10. The block has a weight of 20 lb and is being hoisted at uniform velocity. Determine the angle θ for equilibrium and the force in cord AB .

$\alpha = 0$ since velocity = const.

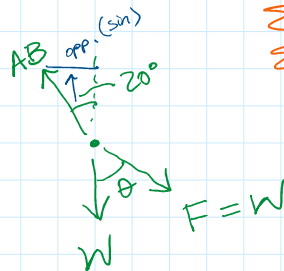
3-10. The block has a weight of 20 lb and is being hoisted at uniform velocity. Determine the angle θ for equilibrium and the force in cord AB.

3-11. Determine the maximum weight W of the block that can be suspended in the position shown if cords AB and CAD can each support a maximum tension of 80 lb. Also, what is the angle θ for equilibrium?



$$a = 0 \text{ since velocity} = \text{const.}$$

FBD @ A



$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$AB_x = -AB \sin(20^\circ)$$

$$AB_y = AB \cos(20^\circ)$$

$$F_x = F \sin \theta$$

$$F_y = F \cos \theta$$

$$\sum F_x = 0$$

$$AB_x + F_x = 0$$

$$-AB \sin(20^\circ) + F \sin \theta = 0$$

$$\textcircled{1} \sin \theta = \frac{AB \sin(20^\circ)}{F}$$

$$AB = \frac{F \sin \theta}{\sin(20^\circ)}$$

$$\sum F_y = 0$$

$$AB_y - F_y - W = 0$$

$$AB \cos(20^\circ) - F \cos \theta - W = 0$$

$$F \cos \theta = AB \cos(20^\circ) - W$$

$$\cos \theta = \frac{AB \cos(20^\circ) - W}{F} = \frac{AB \cos(20^\circ)}{F} - \frac{W}{F}$$

$$\textcircled{2} \cos \theta = \frac{AB \cos(20^\circ)}{F} - 1 \rightarrow \frac{F(\cos \theta + 1)}{\cos(20^\circ)} = AB$$

$$F = 2016$$

$$\frac{F \sin \theta}{\sin(20^\circ)} = \frac{F(\cos \theta + 1)}{\cos(20^\circ)}$$

$$\frac{20}{\sin(20^\circ)} \sin \theta = \frac{20}{\cos(20^\circ)} (\cos \theta + 1)$$

$$58.5 \sin \theta = 21.3 (\cos \theta + 1)$$

$$58.5 \sin \theta = 21.3 \cos \theta + 21.3$$

$$5.85 \sin \theta = 2.13 \cos \theta + 2.13 \quad \text{--- Graph}$$

$$\theta = 40.1^\circ$$